



Treatment of GI Parasitic Infestations — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A patient with amebic colitis is started on metronidazole. What is the cornerstone of this drug's mechanism?

EASY

- A.** It binds parasite β -tubulin and blocks polymerization of microtubules, halting glucose uptake
- B.** It opens glutamate-gated chloride channels in nerve and muscle cells
- C.** A prodrug reduced to a cytotoxic metabolite that damages DNA
- D.** It increases membrane permeability to calcium ions
- E.** It inhibits cholinesterase, paralyzing the organism

2. A patient asks why metronidazole should never be combined with alcohol. What reaction is the concern?

EASY

- A.** A disulfiram-like reaction
- B.** Acute renal failure from tubular crystal formation requiring dialysis
- C.** Serotonin syndrome with agitation, fever and clonus
- D.** Profound, prolonged hypoglycemia needing IV dextrose
- E.** Severe hepatotoxicity requiring transplant evaluation

3. A patient with strongyloidiasis is treated with ivermectin. In which group is this drug contraindicated?

EASY

- A.** Children over age 5
- B.** Pregnant patients
- C.** Patients with peptic ulcer disease
- D.** Patients taking oral contraceptives
- E.** Patients allergic to penicillin

4. A patient with taeniasis (tapeworm infection) cannot tolerate praziquantel. Which drug is the alternative for cestode infections such as taeniasis and diphylobothriasis?

EASY

- A.** Ivermectin
- B.** Pyrantel pamoate
- C.** Iodoquinol
- D.** Chloroquine
- E.** Niclosamide

5. A patient with giardiasis cannot tolerate metronidazole's metallic taste and GI upset. Which single-dose alternative does the lecture describe as possibly superior to metronidazole for this infection?

MEDIUM

- A.** Iodoquinol
- B.** Chloroquine
- C.** Emetine
- D.** Tinidazole
- E.** A benzimidazole

6. A pregnant woman has giardiasis. Which drug can be used to avoid the possible mutagenic effects of metronidazole or tinidazole?

MEDIUM

- A.** Paromomycin
- B.** Nitazoxanide
- C.** Diloxanide furoate
- D.** Emetine
- E.** Chloroquine

7. A patient has intestinal amebiasis and is given iodoquinol together with metronidazole. What does iodoquinol add to the regimen?

MEDIUM

- A. Coverage of anaerobic bacterial co-infection elsewhere in the abdomen
- B. Cyst-clearing activity in the lumen, which metronidazole does not reliably provide
- C. Activity against tissue trophozoites in the intestinal wall and extraintestinal sites
- D. Prevention of the disulfiram-like reaction with alcohol
- E. Inhibition of metronidazole's hepatic metabolism to prolong its therapeutic action

8. A patient with cryptosporidiosis is immunocompetent. Which drug has shown activity in treating this infection and is also used for giardiasis and amebiasis?

MEDIUM

- A. Praziquantel
- B. Pyrantel pamoate
- C. Nitazoxanide
- D. Ivermectin
- E. Niclosamide

9. A patient is treated with paromomycin for intestinal *E. histolytica*. Why is it effective only against the luminal (intestinal) form and not tissue trophozoites?

MEDIUM

- A. It cannot cross the colonic mucosa membrane pumps
- B. It is rapidly metabolized by hepatic CYP450 before reaching tissue
- C. It is destroyed by gastric acid before it can reach the tissue
- D. It binds only to cyst-wall chitin, not trophozoite membranes
- E. It is not significantly absorbed from the gastrointestinal tract

10. A patient with a mixed roundworm and pinworm infection is given a benzimidazole. Which two drugs does the lecture name as the benzimidazoles in wide clinical use?

MEDIUM

- A. Ivermectin and pyrantel pamoate
- B. Praziquantel and niclosamide
- C. Metronidazole and tinidazole
- D. Albendazole and mebendazole
- E. Diloxanide furoate and iodoquinol

11. A patient is given pyrantel pamoate for a hookworm infection. What is its mechanism?

MEDIUM

- A. A depolarizing neuromuscular blocker that releases acetylcholine and inhibits cholinesterase, paralyzing the worm
- B. It inhibits microtubule polymerization by binding β -tubulin
- C. It increases membrane calcium permeability, causing paralysis and death
- D. It opens glutamate-gated chloride channels, hyperpolarizing the worm
- E. It is reduced to a cytotoxic metabolite that fragments the worm's DNA

12. A patient with neurocysticercosis is treated with albendazole. What is its primary therapeutic application, beyond its nematode activity?

MEDIUM

- A. Onchocerciasis (river blindness)
- B. Cestodal infestations such as cysticercosis and hydatid disease, and also microsporidiosis
- C. Amebic liver abscess
- D. Trichomoniasis
- E. Schistosomiasis of all species

13. A patient on metronidazole for trichomoniasis is also taking cimetidine for reflux. What interaction should be anticipated?

MEDIUM

- A. Cimetidine induces metronidazole's metabolism, reducing its therapeutic effect
- B. Cimetidine has no effect on metronidazole; the interaction is with phenobarbitone only
- C. Cimetidine inhibits metronidazole's metabolism, so its dose may need to be decreased
- D. Cimetidine increases renal elimination of metronidazole, requiring a higher dose
- E. Cimetidine causes a disulfiram-like reaction identical to alcohol

14. A patient is prescribed metronidazole while already taking rifampin for tuberculosis. What should be anticipated?

MEDIUM

- A. Rifampin will have no effect since metronidazole is renally cleared unchanged
- B. Rifampin will potentiate metronidazole, requiring a lower dose
- C. Rifampin will cause a disulfiram-like reaction with metronidazole
- D. Rifampin will speed conversion of metronidazole to its active metabolite, increasing toxicity
- E. Rifampin, an enzyme inducer, may reduce metronidazole's therapeutic effect

15. A patient on metronidazole for a mixed anaerobic abdominal infection is also taking warfarin and lithium. Which combination of interactions is correct?

HARD

- A. Reduces warfarin's action by inducing its metabolism, and increases renal elimination of lithium instead
- B. Has no effect on warfarin, but induces lithium metabolism through the same hepatic pathway
- C. Enhances warfarin by inducing hepatic clotting-factor synthesis, a mechanism unrelated to metabolism
- D. Enhances warfarin by inhibiting its metabolism, and can decrease renal lithium elimination, risking toxicity
- E. Decreases lithium absorption in the gut, which would protect against toxicity rather than cause it

16. A clinician is choosing between metronidazole and paromomycin for a patient with amebiasis confined to the intestinal lumen. Which distinguishes the two drugs' spectrum?

HARD

- A. Metronidazole kills trophozoites but not cysts; paromomycin, being luminal-only, acts only in the lumen
- B. Both drugs are equally effective at eradicating cysts from the intestinal lumen
- C. Metronidazole is the luminal-only agent, while paromomycin also treats hepatic abscesses
- D. Iodoquinol, not either of these two drugs, is the only one that treats tissue trophozoites
- E. Paromomycin is contraindicated in amebiasis and is reserved only for cryptosporidiosis

17. A patient with schistosomiasis on praziquantel is also taking phenytoin for seizures, and a second patient on the same drug is taking cimetidine. How does each interacting drug change praziquantel's plasma levels?

HARD

- A. Phenytoin raises praziquantel bioavailability, while cimetidine lowers its plasma concentration instead
- B. Phenytoin lowers praziquantel bioavailability (CYP induction); cimetidine raises its plasma concentration
- C. Both phenytoin and cimetidine lower praziquantel bioavailability by roughly the same amount
- D. Neither drug changes praziquantel levels, since it is not metabolized by any CYP enzyme
- E. Phenytoin has no effect, but cimetidine reduces how well praziquantel is absorbed from food

18. A patient with severe amebic dysentery has failed luminal therapy and needs a drug effective against tissue trophozoites, but the physician is reluctant to use it first-line. Which drug fits, and why is its use limited?

HARD

- A. Diloxanide furoate, said to be ineffective against luminal cysts despite treating tissue disease
- B. Paromomycin, said to be poorly tolerated orally despite treating tissue disease
- C. Emetine or dehydroemetine, whose major toxicity concerns limit routine use
- D. Nitazoxanide, said to be unapproved for amebiasis despite treating tissue disease
- E. Iodoquinol, said to interact with warfarin despite treating tissue disease

19. A patient has both ascariasis and trichuriasis. The lecture notes ivermectin is more effective against one than the other. Which pairing is correct?

HARD

- A. More effective against trichuriasis and hookworm than against ascariasis or enterobiasis
- B. Equally effective against all four of the nematode infections the lecture names
- C. Effective only against hookworm, with no activity against any of the other three worms
- D. More effective against ascariasis and enterobiasis than against trichuriasis or hookworm
- E. Effective only against adult worms of every species, including adult *Onchocerca volvulus*

20. A patient is being treated for a mixed anaerobic infection with metronidazole. Which organism on the lecture's spectrum-of-activity list would NOT be expected to respond?

HARD

- A. *Clostridium difficile*, a listed anaerobic bacterium
- B. *Helicobacter pylori*, a listed anaerobic bacterium
- C. *Bacteroides fragilis*, a listed anaerobic bacterium
- D. *Trichomonas vaginalis*, a listed protozoan
- E. An aerobic Gram-negative rod that is not on the lecture's anaerobe/protozoa list

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	Metronidazole needs reductive activation of its nitro group by susceptible organisms; the reactive metabolite targets DNA.
2	A	EASY	Metronidazole with alcohol can cause a disulfiram-like reaction (nausea, vomiting, sweating, flushing, tachycardia, hypotension).
3	B	EASY	The lecture states ivermectin should not be used in pregnancy.
4	E	EASY	Niclosamide is an alternative to praziquantel for cestode infections (taeniasis, diphyllbothriasis, and others).
5	D	MEDIUM	A single dose of tinidazole may be superior to metronidazole for giardiasis.
6	A	MEDIUM	Paromomycin can be used to treat pregnant women with giardiasis to avoid the possible mutagenic effects of the other drugs.
7	B	MEDIUM	Iodoquinol is effective against luminal trophozoite and cyst forms, but not tissue or extraintestinal trophozoites — it is used together with metronidazole for this reason.
8	C	MEDIUM	Nitazoxanide is used for cryptosporidiosis, giardiasis, and amebiasis, and also has activity against intestinal helminths.
9	E	MEDIUM	Paromomycin is only effective against the luminal (intestinal) forms of <i>E. histolytica</i> because it is not significantly absorbed from the GI tract.
10	D	MEDIUM	Although many benzimidazoles have been synthesized, only albendazole and mebendazole are in wide clinical use.
11	A	MEDIUM	Pyrantel pamoate acts as a depolarizing, neuromuscular-blocking agent — it releases acetylcholine and inhibits cholinesterase, paralyzing the worm so it is expelled.
12	B	MEDIUM	Albendazole's primary therapeutic application is in cestodal infestations (cysticercosis, hydatid disease); it is also effective in microsporidiosis, a fungal infection — onchocerciasis is ivermectin's domain.
13	C	MEDIUM	Cimetidine (an enzyme inhibitor) can reduce metronidazole metabolism, so its dose may need to be decreased.
14	E	MEDIUM	Phenobarbitone and rifampin are enzyme inducers that may reduce metronidazole's therapeutic effect.
15	D	HARD	Metronidazole enhances warfarin's action by inhibiting its metabolism, and can decrease renal elimination of lithium, precipitating toxicity.
16	A	HARD	Metronidazole kills trophozoites but not cysts; paromomycin acts only in the lumen since it is not significantly absorbed from the GI tract.
17	B	HARD	Praziquantel bioavailability is reduced by hepatic CYP inducers (phenytoin, carbamazepine, phenobarbital) and corticosteroids, but its plasma concentration increases with cimetidine.
18	C	HARD	Emetine and dehydroemetine are effective against tissue trophozoites of <i>E. histolytica</i> , but major toxicity concerns limit their use; diloxanide furoate, by contrast, is the luminal-only agent.
19	D	HARD	Ivermectin is more effective in ascariasis and enterobiasis than in trichuriasis or hookworm infection, and kills <i>Onchocerca microfilariae</i> but not adult worms.
20	E	HARD	Metronidazole's spectrum covers the listed protozoa (<i>E. histolytica</i> , <i>Giardia lamblia</i> , <i>Trichomonas vaginalis</i>) and anaerobic bacteria (<i>C. difficile</i> , <i>C. perfringens</i> , <i>H. pylori</i> , <i>fusobacterium</i> , <i>B. fragilis</i> , <i>Gardnerella vaginalis</i>); an aerobic organism outside this list is not covered.